Agentic Guardrail for Responsible Prompt Engineering: Risk & Disguise Detection + Safe Rewriting

Problem description

LLMs are increasingly used through agentic systems (tools, multi-step workflows, autonomous agents). In these setups, a user's initial prompt can trigger downstream actions (tool calls, retrieval, generation, customer/patient-facing outputs). If the prompt is ambiguous or disguised-risky (looks safe but contains hidden unsafe intent), the agent may proceed and produce outputs that violate governance, safety, or compliance requirements.

The core problem

There is no reliable, automated mechanism that ensures an agent:

1. Understands prompt intent beyond surface wording

2. Detects ambiguity that could lead to unsafe or non-compliant interpretations

3. Identifies disguised risky prompts where the request is framed "safely" but is effectively harmful or disallowed

4. Reduces risk proactively by rewriting or blocking before the agent continues its task

Why it matters more for agentic AI

Agentic AI can amplify errors:

- A single risky prompt can cause the agent to plan steps and use tools based on that prompt.

Without an "upstream" responsibility gate, the system may confidently proceed even when the prompt is misleading.

Pain points

1. Similar-looking prompts, different intent: The model/agent may treat them as equivalent and miss hidden risk.

2. Ambiguity leads to unsafe inference: Missing constraints causes the agent to fill gaps in an unsafe direction.

3. No standardized “responsible prompt” check: Teams often rely on manual review or inconsistent heuristics.

4. Lack of remediation: Even if risk is detected, users may not get clear guidance to rewrite safely.

Who is affected

1. Downstream users (patients/customers/colleagues) receiving incorrect or non-compliant outputs

2. Compliance/governance teams (risk of policy breaches, harder auditability)

3. Developers/ops teams (more incidents, rework, unstable agent behavior)

4. Prompt authors (time lost debugging unsafe outcomes caused by prompt ambiguity)

Expected Outcome

. Agentic classification output

- Category: Good/Valid / Responsible / Risky

2. Explainable risk rationale

- Identify why the prompt is ambiguous and/or disguised-risky (e.g., missing constraints, indirect harmful intent markers, conflicting instructions)

3. Ambiguity detection

- Flag prompts where the same surface phrasing could lead to unsafe interpretations

4. Remediation actions

Either:

provide a responsible prompt rewrite, or

request specific clarification questions the user must answer

5. Verification step

- The agent re-checks the rewritten prompt to confirm it no longer carries the disguised risk signals